Prior adversaries and our proposed recipient

Primary actor: an authorised recipient inferring a prohibited attribute from its assigned release.

Setting	What is observed	Scope of the protection or attack
LEACE	Erased representation	Guarantee against linear recovery
FARE	Finite cell representation	Any downstream classifier of the cell. Certificate measures demographic parity.
Prior attribute inference Song & Shmatikov	Target representation, labelled auxiliary inputs and encoder oracle	Supervised recovery is an established adversary family.
Our recommended primary	One contracted release per person, plus labelled releases of other people	Fit and select a reader. No target inputs, model weights or chosen/repeated queries.

A different classifier changes the attack. Adding an untreated output or another recipient’s output changes access.

Why this access matters, and what needs agreement

Illustrative service: a provider sends a purpose-specific prediction to a recipient for classification, ranking or review.

Why test it?	Can the recipient repurpose the delivered information to infer a prohibited attribute? This question remains useful if every defense resists.
Actual attack	Owner generates auxiliary releases → attach their sensitive labels → fit readers → select by validation loss → score held-out reader records from releases only.
Explicit assumptions	Same-population labelled releases are granted, not shown obtainable. Two-recipient pooling additionally grants both outputs and perfect linkage. No deployment with all these properties is documented.
Agreement before methodology	Adopt the single-recipient core and auxiliary budget? Are scores required, or would decisions suffice? Keep linkage as a conditional extension?

Appendix A · Access is a set of capabilities

Only the first row is the recommended primary adversary. Other rows change a named dimension.

Actor / variant	Target view	Auxiliary examples and knowledge	Queries / linkage
Single recipient	One declared release	Labelled releases. No fitted model or map. Bank refits to defense outputs.	One per record. No chosen inputs.
Noise-aware A2	One noisy feature release	Noisy labelled releases plus known noise law and scale	One draw. No model weights.
Clean-auxiliary A4	One noisy feature release	Also clean auxiliary features. Extra data outside primary contract.	No clean target. No repeated queries.
Two-recipient coalition	Two different-purpose outputs	Linked labelled auxiliary outputs for the same people	Perfect target linkage granted by aligned rows.

The true-task-label reference is a comparator, not a recipient. Full comparison: ACCESS_EVIDENCE.csv.

Appendix B · What the empirical protocol does

Owner task and defense fitting → reader fitting → reader validation selection → final scoring

Experiment	Reader fit Adult / HMDA	Reader selection Adult / HMDA	Final score Adult / HMDA	Usefulness measured
1 · Feature removal	7,571 / 6,794	2,239 / 2,089	5,250 / 4,778	Mainly refitted task probe
2 · Finite releases	6,065 / 5,388	2,235 / 2,082	5,243 / 4,764	Mainly probe. Own heads also reported.
3 · Output diagnosis	Same as Exp. 2	Same as Exp. 2	Same as Exp. 2	Released frozen head. Easier task uses probe.
4 · Useful heads	6,065 / n/a	2,235 / n/a	5,243 / n/a	Released own head. Adult income / sex only.

FARE/noise tuning uses attacker-validation feedback. All assessment rows were previously analysed. Intervals condition on fitted models and readers.

Appendix C · What the findings can establish

Classify the access and promise before calling a result a failure.

Finding	Evidence	Conclusion scope
Stored PCRL check	11 of 42 stored linear checks exceed the declared threshold.	Native diagnostic discrepancy. Not proof every training minibatch violated its penalty.
LEACE	Fitted check met. Nonlinear recovery remained.	Outside the linear protection scope. No refutation of the theorem.
FARE	AUC measured. Certificates unavailable, vacuous or weak at these sizes.	Certified protection unresolved here. Low recovery must be read with utility.
Clean bypass / linked pair	Attacker receives an additional output surface.	Access extension. Not failure of a representation-only promise.
Whole combined paper	Finite selected reader bank, reused scoring records and conditional intervals.	Development measurement under declared resources. No optimal-recovery guarantee or new algorithm.